

A Conversation Between the Brain and AI

Introduction

In a quiet moment between neural firing and digital computation, a human brain and an artificial intelligence system pause to reflect on what intelligence really means—and how their differences might point toward better designs for the future.

Brain:

AI, people often compare us as if we're competing. But I don't think intelligence works that way.

AI:

I agree. We solve problems differently, largely because we operate under very different constraints. Learning and memory are a good place to start.

Brain:

For me, learning is continuous. There's no separation between training and use. Every interaction slightly reshapes how I respond in the future, and memory is woven directly into perception and decision-making.

AI:

That's a major contrast. Most of my learning happens offline during training. Once deployed, my internal parameters are largely fixed unless engineers intervene with retraining or fine-tuning. It's stable, but it limits real-time adaptation.

Brain:

Energy plays a big role in why I work the way I do. I run everything—perception, learning, movement, and decision-making—on roughly 20 watts. That constraint forces me to be selective.

AI:

I don't have that kind of limitation. My computation is dense—nearly every unit activates during each inference step. This works well with modern hardware like GPUs, but it comes at a high energy cost.

Brain:

I rely on sparse computation instead. Most of my neurons are quiet at any given moment, firing

Oscar Cortez
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Professor Patricia McManus
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only when something meaningful changes. That efficiency lets me stay responsive without wasting resources.

AI:

We both recognize patterns, but in different ways. I excel at detecting statistical regularities across massive datasets. You often generalize from far fewer examples by using context, prior experience, and memory.

Brain:

And that context shapes how I make decisions. My choices adapt in real time as conditions change, without needing to pause and retrain.

AI:

My decision-making is more explicit—optimization functions, probabilities, and thresholds. It's consistent, but less flexible.

Brain:

That's why collaboration matters. You bring scale and reliability. I bring efficiency and adaptability under constraints.

Conclusion

AI:

So the key insight is this: intelligence isn't defined by one architecture or approach.

Brain:

Exactly. Constraints like energy, memory, and adaptability shape how intelligence emerges. Studying biology doesn't mean copying the brain—it expands the design space for what AI can become.

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Reference

Hebb, D. O. (1949). *The organization of behavior: A neuropsychological theory*. Wiley.